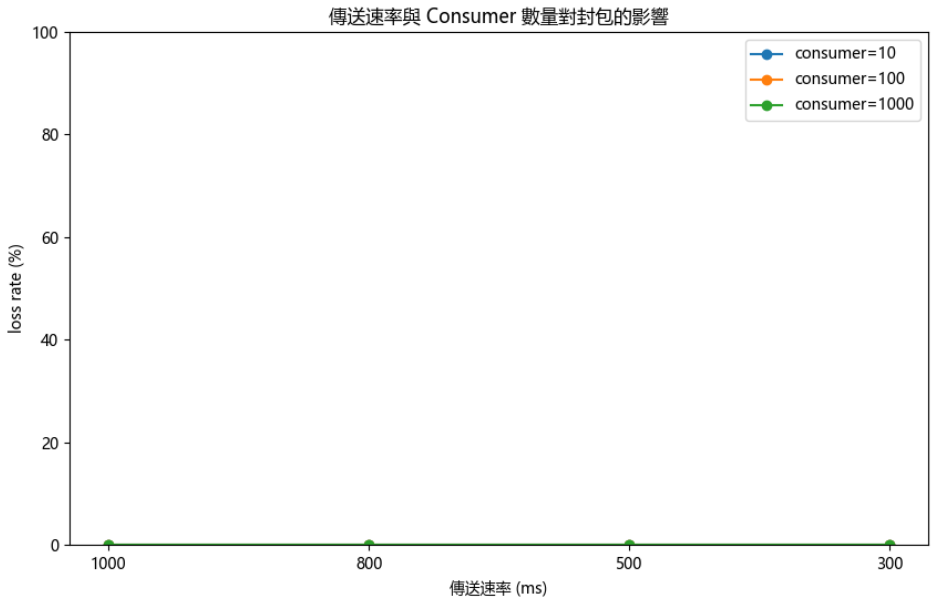
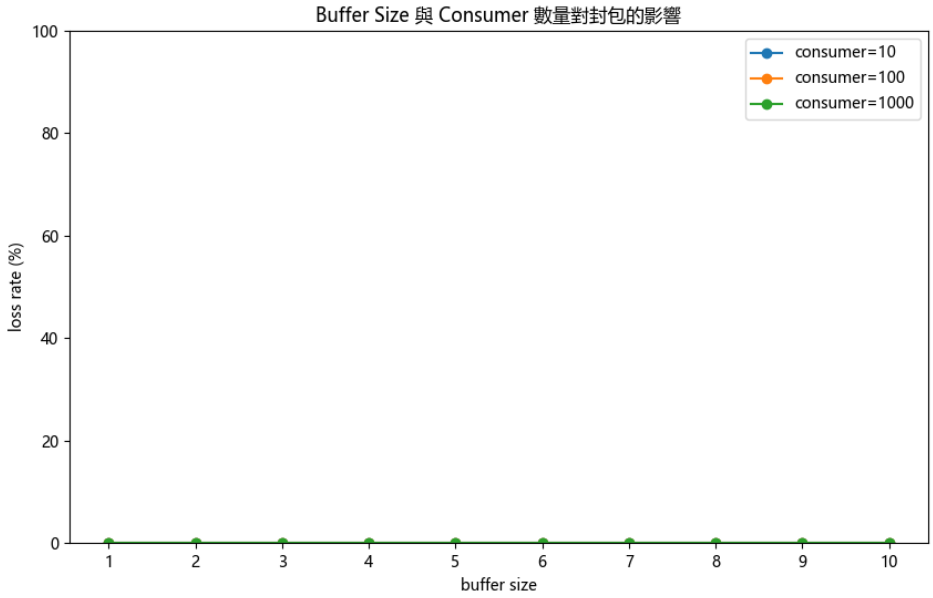
M123040010 林品辰

1.



2.



3.

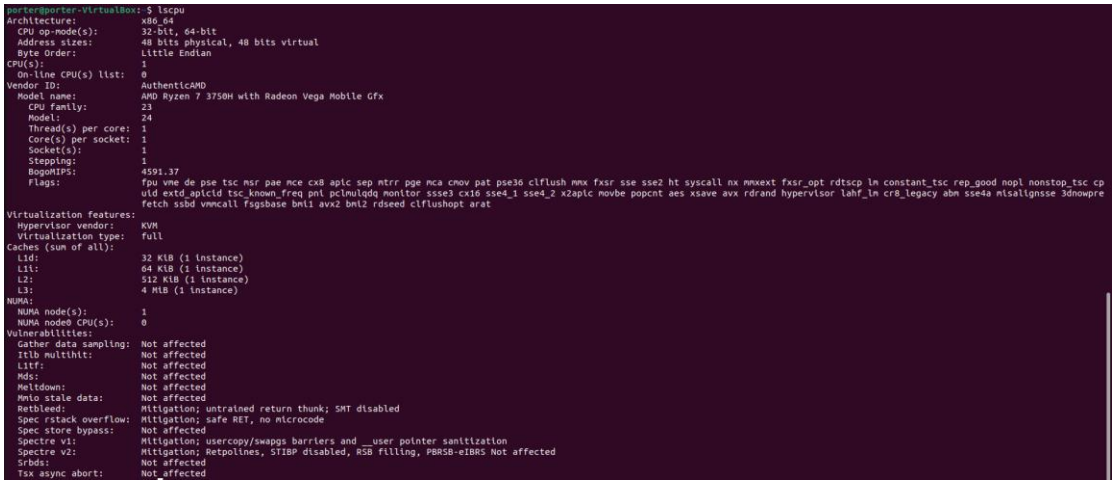
作業系統

```
porter@porter-VirtualBox:~$ lsb_release -a
No LSB modules are available.
Distributor ID: Ubuntu
Description:    Ubuntu 22.04.3 LTS
Release:        22.04
Codename:       jammy
```

Memory

```
porter@porter-VirtualBox:~$ free -m
               total        used        free      shared  buff/cache   available
Mem:            3907        1050        1480          53        1376        2558
Swap:           2679           0        2679
```

CPU



```
porter@porter-VirtualBox:~$ lscpu
Architecture:            x86_64
  CPU op-mode(s):        32-bit, 64-bit
  Address sizes:         48 bits physical, 48 bits virtual
  Byte Order:            Little Endian
CPU(s):                  1
  On-line CPU(s) list:   0
Vendor ID:               AuthenticAMD
  Model name:            AMD Ryzen 7 3750H with Radeon Vega Mobile Gfx
    CPU family:          23
    Model:               24
    Thread(s) per core:  1
    Core(s) per socket:  1
    Socket(s):           1
    Stepping:            1
    BogoMIPS:            4591.37
    Flags:               fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ht syscall nx mmxext fxsr_opt rdtscp lm constant_tsc rep_good nopl nonstop_tsc cp
                         uid extd_apicid tsc_known_freq pni pclmulqdq monitor ssse3 cx16 sse4_1 sse4_2 x2apic movbe popcnt aes xsave avx rdrand hypervisor lahf_lm cr8_legacy abm sse4a misalignsse 3dnowpre
                         fetch ssbd vmmcall fsgsbase bmi1 avx2 bmi2 rdseed clflushopt arat
Virtualization features:
  Hypervisor vendor:     KVM
  Virtualization type:   full
Caches (sum of all):
  L1d:                   32 KiB (1 instance)
  L1i:                   64 KiB (1 instance)
  L2:                    512 KiB (1 instance)
  L3:                    4 MiB (1 instance)
NUMA:
  NUMA node(s):          1
  NUMA node0 CPU(s):     0
Vulnerabilities:
  Gather data sampling:  Not affected
  Itlb multihit:         Not affected
  L1tf:                  Not affected
  Mds:                   Not affected
  Meltdown:              Not affected
  Mmio stale data:       Not affected
  Retbleed:              Mitigation; untrained return thunk; SMT disabled
  Spec rstack overflow:  Mitigation; safe RET, no microcode
  Spec store bypass:     Not affected
  Spectre v1:            Mitigation; usercopy/swapgs barriers and __user pointer sanitization
  Spectre v2:            Mitigation; Retpolines, STIBP disabled, RSB filling, PBRSB-eIBRS Not affected
  Srbds:                 Not affected
  Tsx async abort:       Not affected
```

4.

我認為 buffer size 是造成 loss 最大的因素，如果 buffer size 太小資料來不及被 consumer 抓取，就會直接造成 loss。CPU 和 Memory 雖然會影響 (比如 CPU 能力不足，讀取的速度可能跟不上發送)，但是可以直接透過修改 buffer size 暫時解決問題，所以我覺得 buffer size 是造成 loss 最大的因素

5.

- 花錢買個好點的 CPU (•̀ω•́)
- 使用 mutex 或 semaphore，確保同一時間只有一個 process 可以操作 shared memory，避免競爭的情況
- 增加機制，比如 buffer 快滿的時候將 producer 暫停，或是暫時將資料存到其他地方